EXPERIMENT - 12

AIM :

To implement a Tic Tac Toe game using Python for two players.

ALGORITHM:

1. Start the program and initialize a 3 × 3 board.
2. Display the board.
3. Get the position from the current player.
4. Check whether the position is valid and empty.
5. Place the player's symbol on the board.
6. Check for a win or a draw.
7. Repeat until the game ends.
8. Display the result and stop.

PROGRAM :

untitled (3.14.8)

File Edit Format Run Options Window Help

```
board = [' '] * 9

def display():
    for i in range(0, 9, 3):
        print(board[i], '|', board[i+1], '|', board[i+2])
        if i < 6:
            print('--+---+--')

def check_win(p):
    wins = [
        (0,1,2), (3,4,5), (6,7,8),
        (0,3,6), (1,4,7), (2,5,8),
        (0,4,8), (2,4,6)
    ]
    return any(all(board[i] == p for i in line)
               for line in wins)

player = 'X'

while ' ' in board:
    display()
    pos = int(input(
        f"Player {player}, enter position (1-9): "
    )) - 1

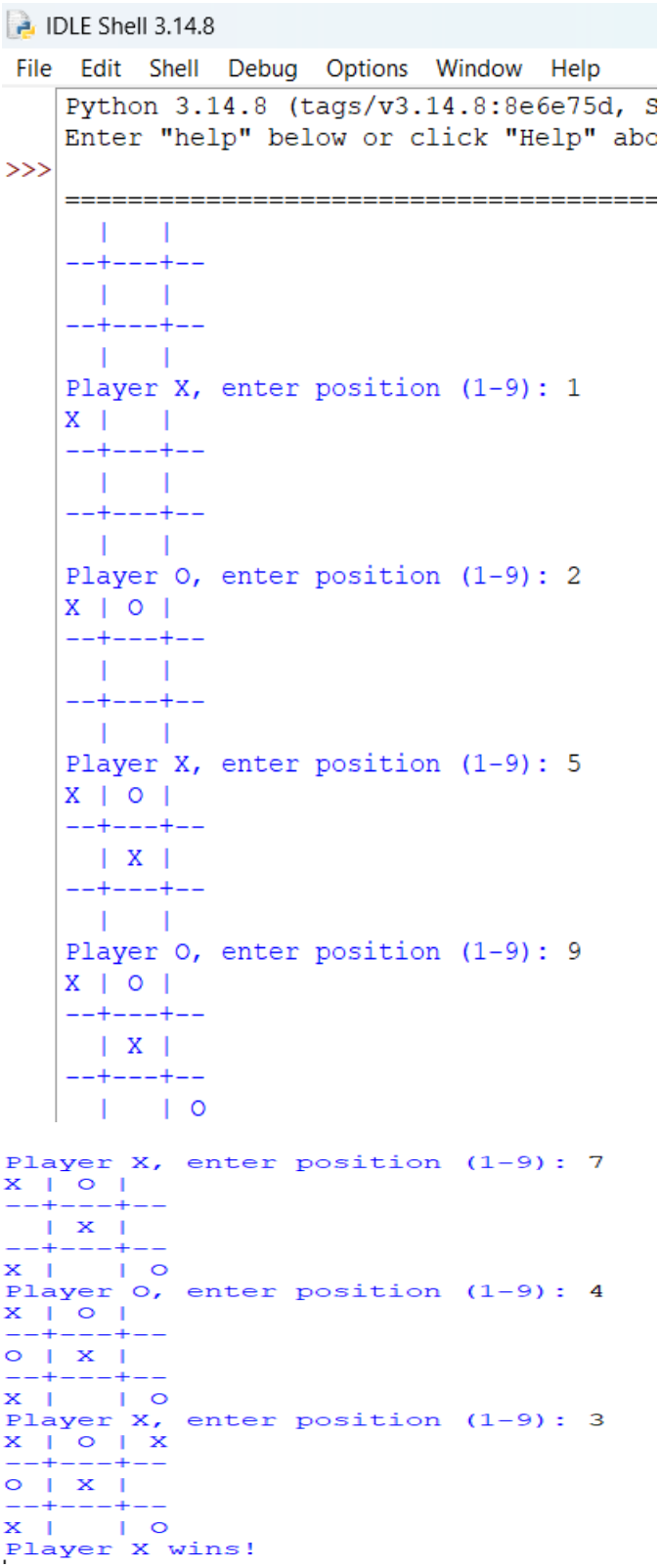
    if pos < 0 or pos > 8 or board[pos] != ' ':
        print("Invalid position! Try again.")
        continue

    board[pos] = player

    if check_win(player):
        display()
        print("Player", player, "wins!")
        break

    player = 'O' if player == 'X' else 'X'
else:
    display()
    print("Match Draw!")
```

OUTPUT:



RESULT:

Thus, the Tic Tac Toe game was implemented successfully, and the winner was identified.